Topic: Phase portraits for complex Eigenvalues

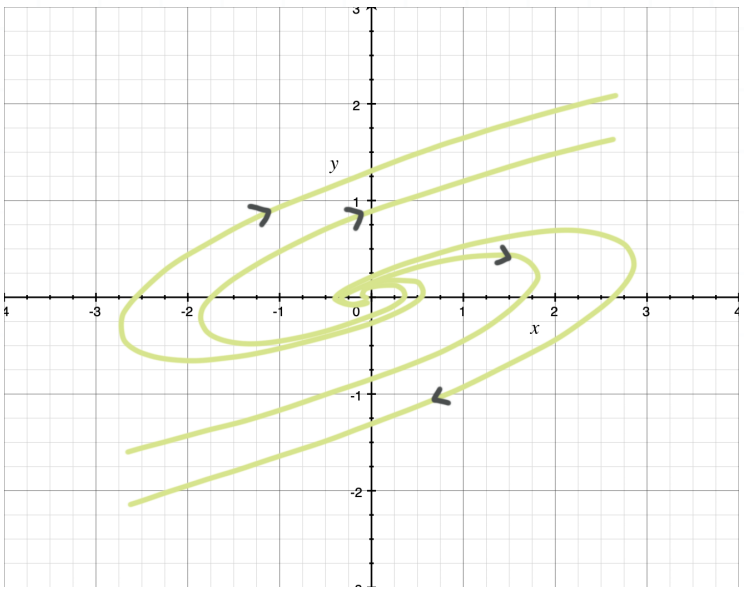
Question: Sketch the phase portrait of the system.

$$x_1' = x_1 - 8x_2$$

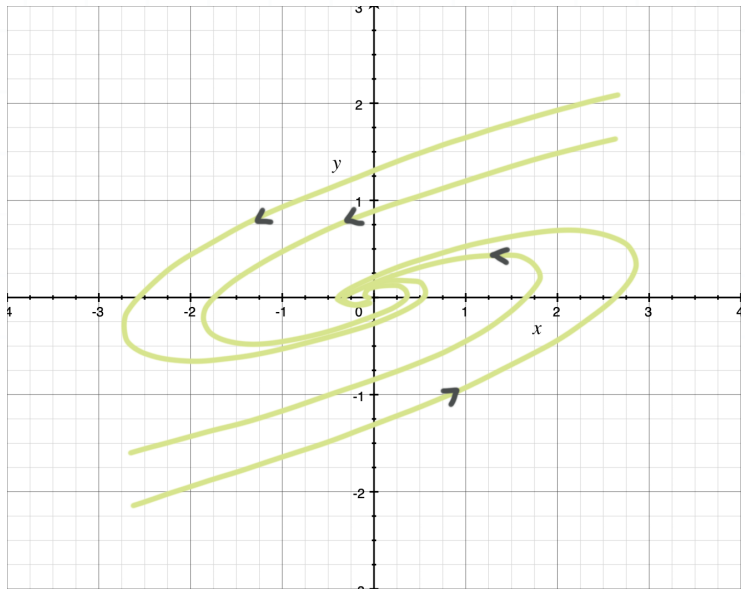
$$x_2' = x_1 - 3x_2$$

Answer choices:

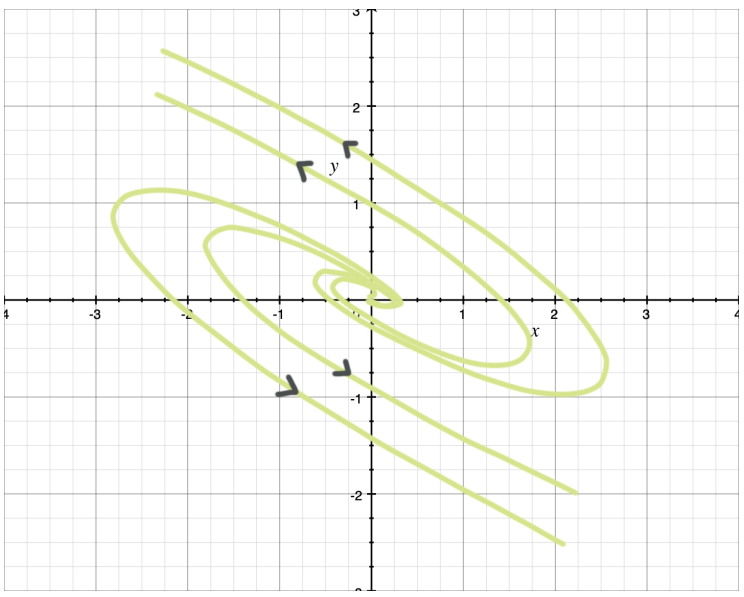
A



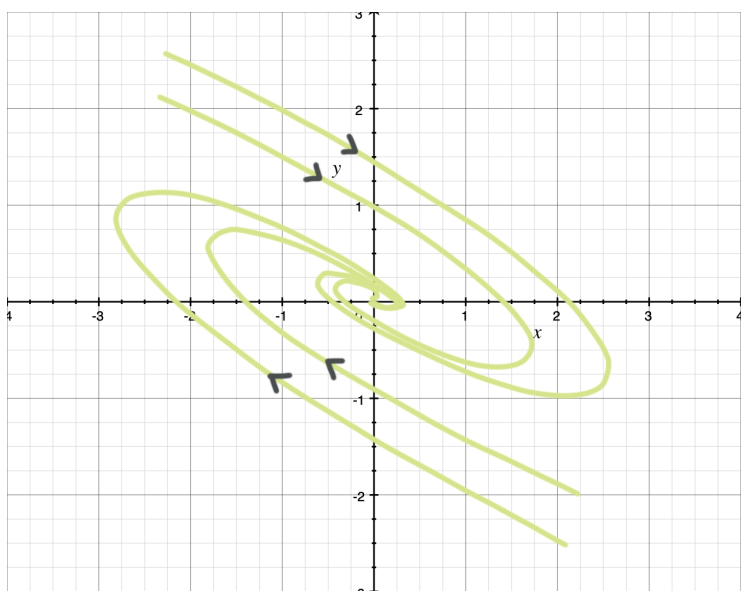
B



C



D



© Krista King Math

Solution: B

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 1 & -8 \\ 1 & -3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & -8 \\ 1 & -3 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(1 - \lambda)(-3 - \lambda) - (-8)(1) = 0$$

$$\lambda^2 + 2\lambda + 5 = 0$$

$$(\lambda + 1)^2 = -4$$

$$\lambda = -1 \pm 2i$$

These are Eigenvalues with a negative real part, $\alpha = -1 < 0$, which means we're dealing with an asymptotically stable attractor spiral that attracts all trajectories.

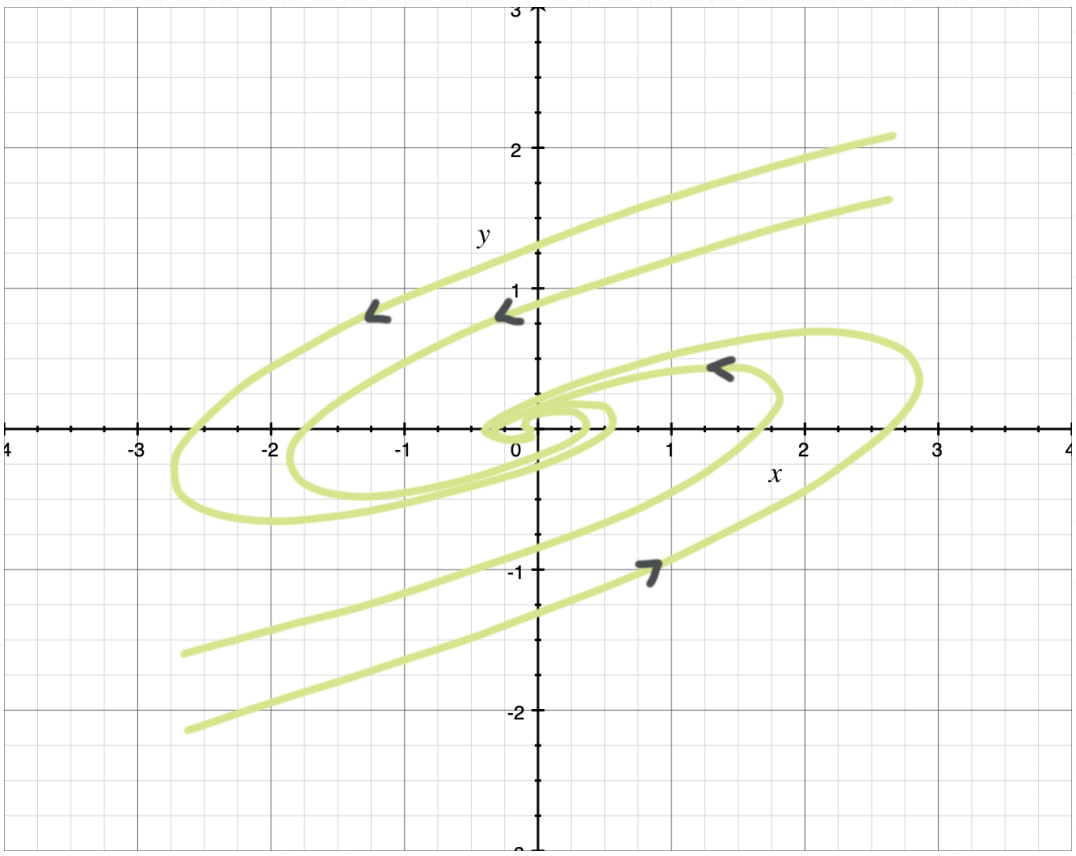
If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

$$\vec{x}' = \begin{bmatrix} 1 & -8 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

© Krista King Math

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(1,1)$ (into the first quadrant), which means that the phase portrait must look something like



© Krista King Math

Topic: Phase portraits for complex Eigenvalues

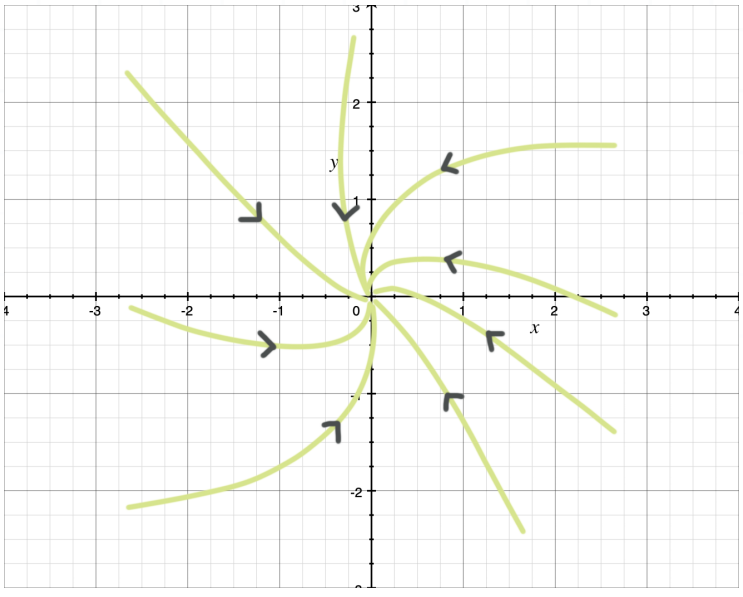
Question: Sketch the phase portrait of the system.

$$x_1' = 5x_1 + x_2$$

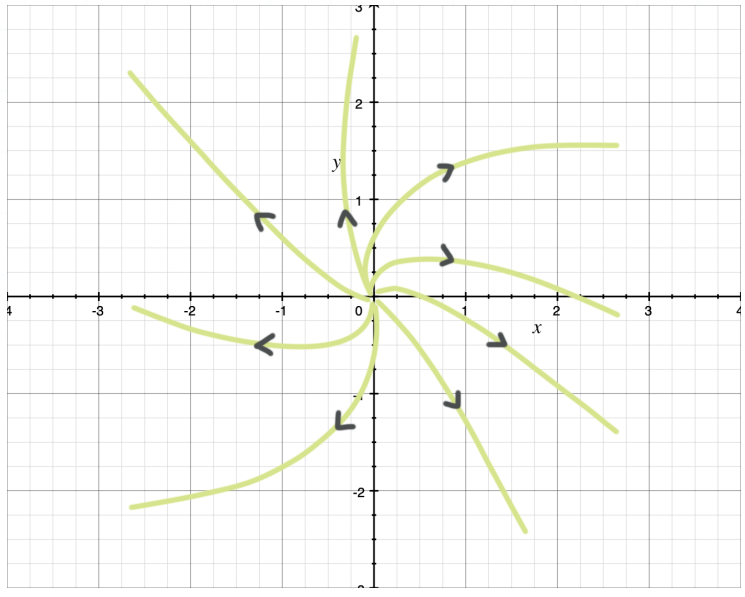
$$x_2' = -2x_1 + 3x_2$$

Answer choices:

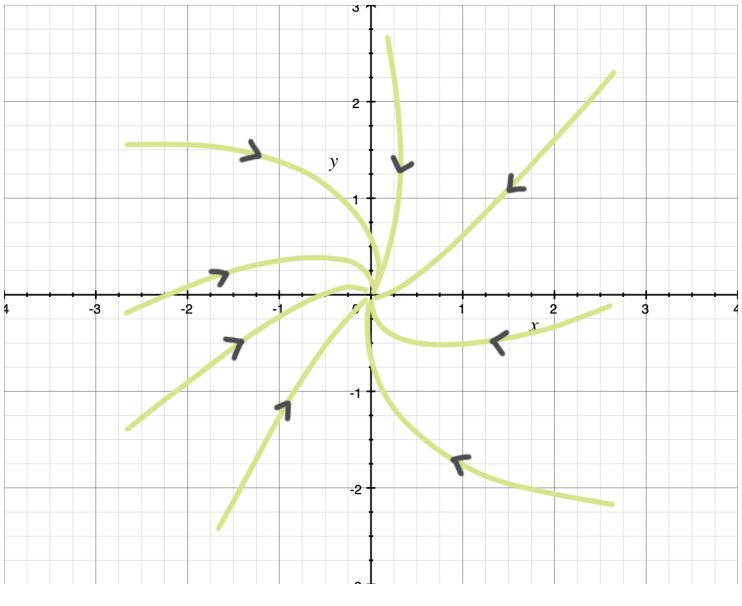
A



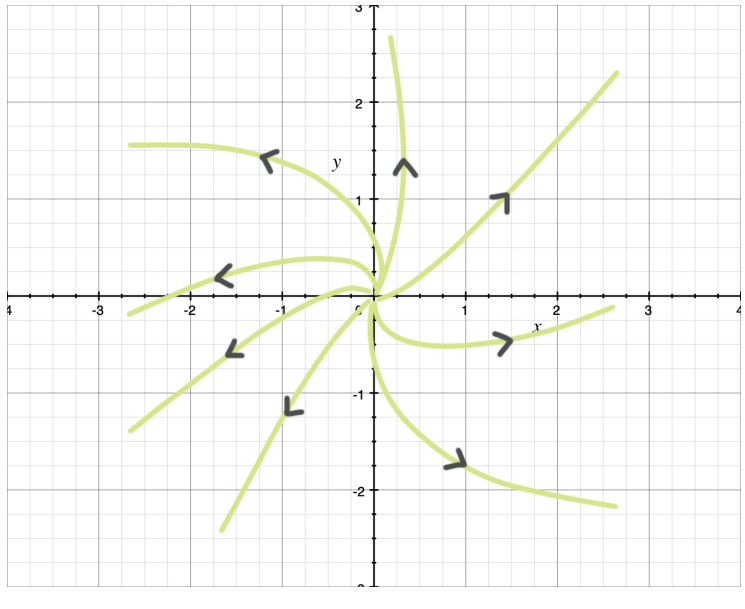
B



C



D



© Krista King Math

Solution: B

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 5 & 1 \\ -2 & 3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 5 - \lambda & 1 \\ -2 & 3 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(5 - \lambda)(3 - \lambda) - (1)(-2) = 0$$

$$\lambda^2 - 8\lambda + 17 = 0$$

$$(\lambda - 4)^2 = -1$$

$$\lambda = 4 \pm 1i$$

These are Eigenvalues with a positive real part, $\alpha = 4 > 0$, which means we're dealing with an unstable repeller spiral that repels all trajectories.

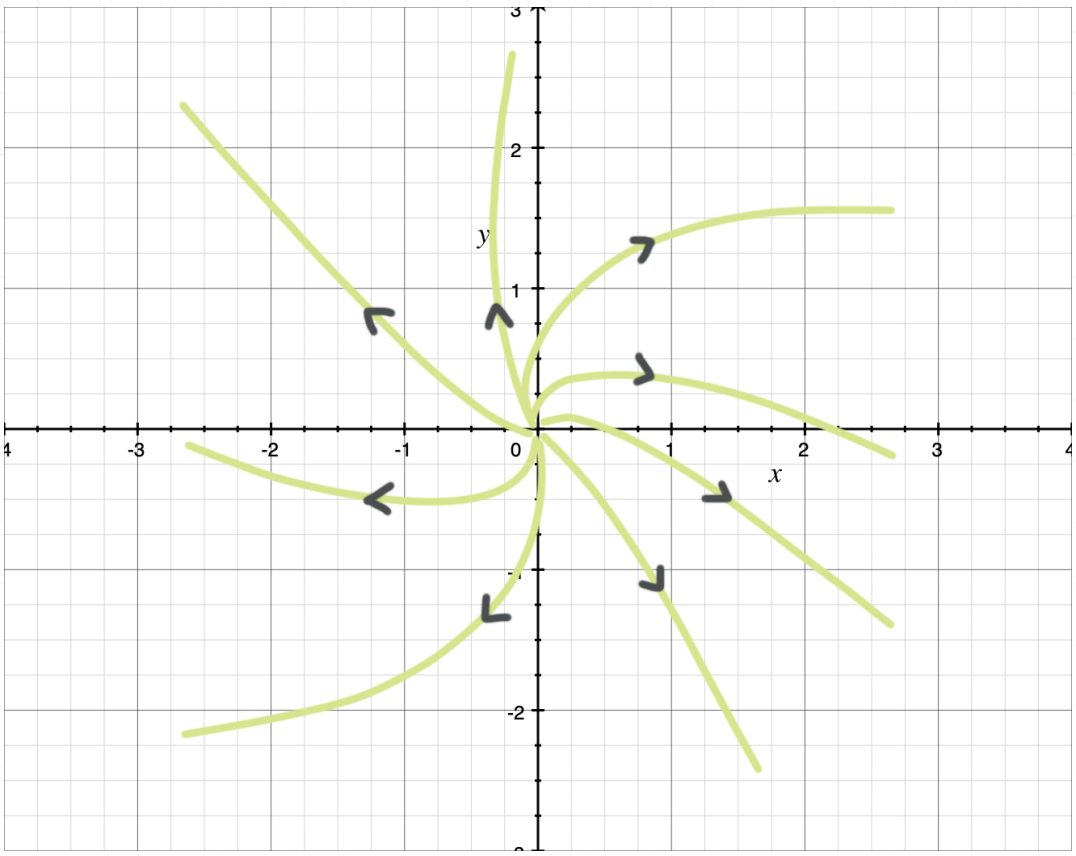
If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

$$\vec{x}' = \begin{bmatrix} 5 & 1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

© Krista King Math

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(5, -2)$ (into the fourth quadrant), which means that the phase portrait must look something like



© Krista King Math

Topic: Phase portraits for complex Eigenvalues

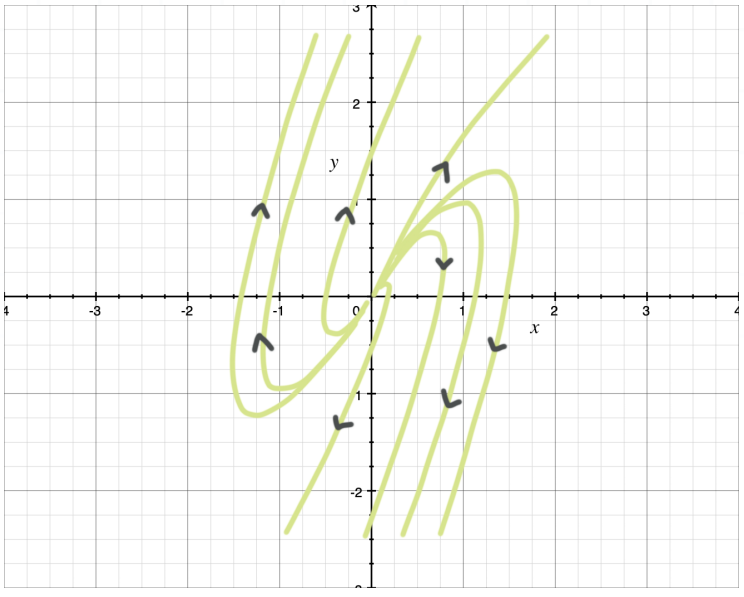
Question: Sketch the phase portrait of the system.

$$x_1' = 4x_1 - 5x_2$$

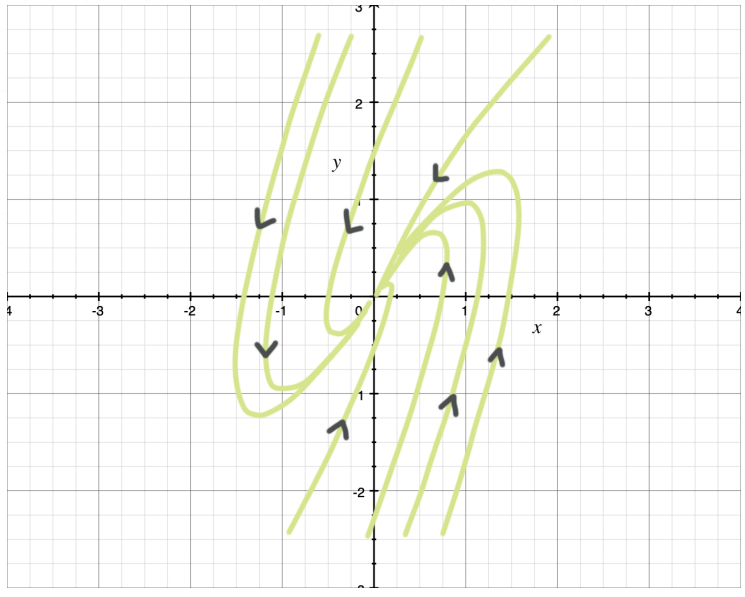
$$x_2' = 5x_1 - 4x_2$$

Answer choices:

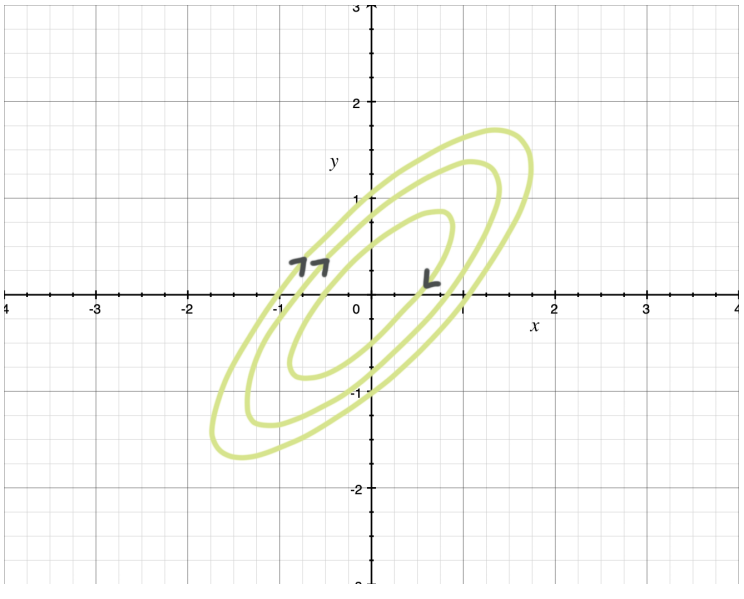
A



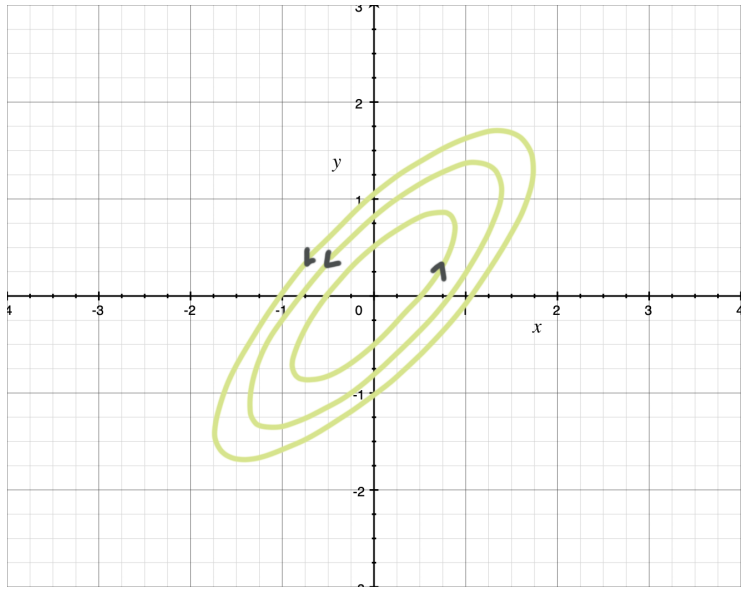
B



C



D



© Krista King Math

Solution: D

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 4 & -5 \\ 5 & -4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & -5 \\ 5 & -4 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(4 - \lambda)(-4 - \lambda) - (-5)(5) = 0$$

$$\lambda^2 = -9$$

$$\lambda = \pm 3i$$

These are Eigenvalues with no real part, $\alpha = 0$, which means we're dealing with a stable center that neither repels nor attracts trajectories.

If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

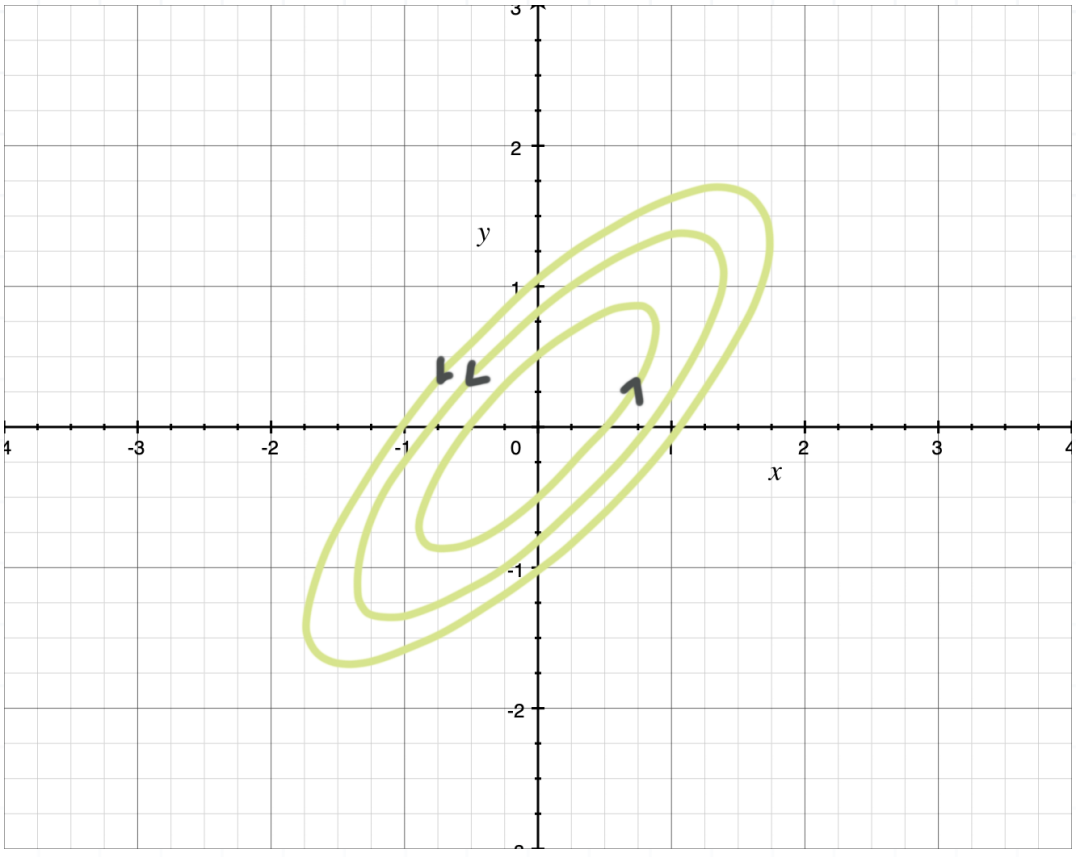
$$\vec{x}' = \begin{bmatrix} 4 & -5 \\ 5 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(4,5)$ (into the first quadrant), which means that the phase portrait must look something like

© Krista King Math





© Krista King Math